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Mott–Schottky Analysis and Impedance Spectroscopy of TiO₂/6T and ZnO/6T devicesAnahita Mani,^{*,†,‡} Carolien Huisman,[†] Albert Goossens,^{†,‡} and Joop Schoonman[†]*Laboratory for Inorganic Chemistry, Delft University of Technology, P.O. Box 5045, 2600 GA Delft, The Netherlands and Dutch Polymer Institute (DPI), P.O. Box 902, 5600 AX Eindhoven, The Netherlands**Received: February 16, 2008; Revised Manuscript Received: June 04, 2008*

Schottky junctions have been realized by evaporating gold spots on top of sexithiophen (6T), which is deposited on TiO₂ or ZnO with e-beam and spray pyrolysis. Using Mott–Schottky analysis of 6T/TiO₂ and 6T/ZnO devices acceptor densities of 4.5×10^{16} and $3.7 \times 10^{16} \text{ cm}^{-3}$ are obtained, respectively. For 6T/TiO₂ deposited with the e-beam evaporation a conductivity of $9 \times 10^{-8} \text{ S cm}^{-1}$ and a charge carrier mobility of $1.2 \times 10^{-5} \text{ cm}^2/\text{V s}$ is found. Impedance spectroscopy is used to model the sample response in detail in terms of resistances and capacitances. An equivalent circuit is derived from the impedance measurements. The high-frequency data are analyzed in terms of the space-charge capacitance. In these frequencies shallow acceptor states dominate the heterojunction time constant. The high-frequency RC time constant is $8 \mu\text{s}$. Deep acceptor states are represented by a resistance and a CPE connected in series. The equivalent circuit is validated in the potential range (from -1.2 to 0.8 V) for 6T/ZnO obtained with spray pyrolysis.

Introduction

Conjugated oligomer semiconductors have potential for application in new optoelectronic devices, such as organic solar cells. The performance of these devices depends strongly on the properties of the semiconductor materials. Compared to polymers, the structure of oligomers is better defined, which gives access to fine tuning of their properties. Films of oligomers can be obtained with vacuum deposition, offering an excellent reproducibility of the deposition conditions and hence film properties. Thiophene–oligomer films are polycrystalline and behave as p-type semiconductors. They exhibit good device characteristics if used as active layers in electronic devices such as Schottky diodes,^{1,2} electroluminescent devices,³ and thin-film transistors (TFTs).^{4,5} Sexithiophene (6T) has a band gap of 2.3 eV (Figure 1) and a strong optical absorption in the range of 400–550 nm. Hole mobilities of approximately 1 and 0.5 cm²/Vs for field-effect transistors (FET) employing polycrystalline films of pentacene^{6,7} and 6T,⁸ respectively, have been reported.

Dye-sensitized nanostructured metal oxide films based on dithiocyanato-bis-(4,4′-dicarboxy 2,2′-bipyridine)ruthenium(II), Ru(dcbpy)₂(NCS)₂, adsorbed on nanoporous TiO₂ thin films (Grätzel cells) are investigated intensively. They show solar energy to electricity conversion of $\sim 11\%$ under 1 sun (AM 1.5).^{9–12} Recently, we reported that 6T in TCO/TiO₂/6T/Au heterojunctions act as a sensitizer for TiO₂ and shows good charge-transfer characteristics.¹³ However, detailed impedance spectroscopy investigations of these heterojunctions are lacking, and therefore, their optoelectrical behavior is not yet understood well.

In addition to TiO₂, ZnO also can be used as an electron acceptor in sensitized solar cells. Gerischer et al. used ZnO for the semiconductor electrode and organic dyes such as 9-phenylxanthene as the photosensitizer.^{14,15}

Cells based on nanostructured ZnO exhibit lower efficiencies compared to cells based on TiO₂. Several explanations have

been suggested in the literature, including poor surface coverage and weak electronic coupling between the dye and ZnO.^{16–19} Nanostructured ZnO exhibits different and often improved properties compared to the bulk phase, and hence, the study of electrical properties of this nanostructured material is important.^{20,21}

The electrical properties of metal oxide/organic composites in solar cells are strongly influenced by highly resistive or conductive grain boundaries. Recently, there has been considerable interest in the study of the electrical properties and charge transfer in nanostructured p-n-junctions. Impedance spectroscopy, which is sensitive to small changes in solid-state nanostructured devices, is a powerful technique for studying electron transport and charge recombination in nanocrystalline films.

In this study, impedance spectroscopy is focused on the charge-transport process in Au/6T/TiO₂/TCO/glass and Au/6T/ZnO/TCO/glass devices along with Mott–Schottky analysis of 6T/TiO₂ and 6T/ZnO interfaces as well as fitting of impedance data to obtain a well-determined electrical equivalent circuit. The equivalent circuit elements are interpreted including both the space charge (SC) layer and the charge trapping.

Experimental Aspects

Sample Preparation. ZnO and TiO₂ films are deposited with spray-pyrolysis deposition (SPD). The following procedure is used for ZnO deposition using a solution of Zn(CH₃COO)₂·2H₂O in ethanol (99.99%) including 14 mL of acetyl acetate. The substrate is kept at 350 °C during the deposition in air. The optimal quality of the dense films is obtained for a waiting time between spraying steps of 50 s. Dense 100 nm thick anatase TiO₂ films are deposited using a solution of 54 mL of ethanol (99.99%), 3.6 mL of acetyl acetate, and 2.4 mL of titanium tetraisopropoxide (TTIP, 97%). The precursors are mixed and sprayed on a heated TCO glass (transparent conducting oxide; SnO₂:F coated glass, TEC 20/2.5 mm, Libbey Owens Ford or indium–tin oxide, ITO) substrate. The substrate surface is maintained at a constant temperature of 350 °C. Only the gas flow and waiting time between every spraying step are varied. The substrates are

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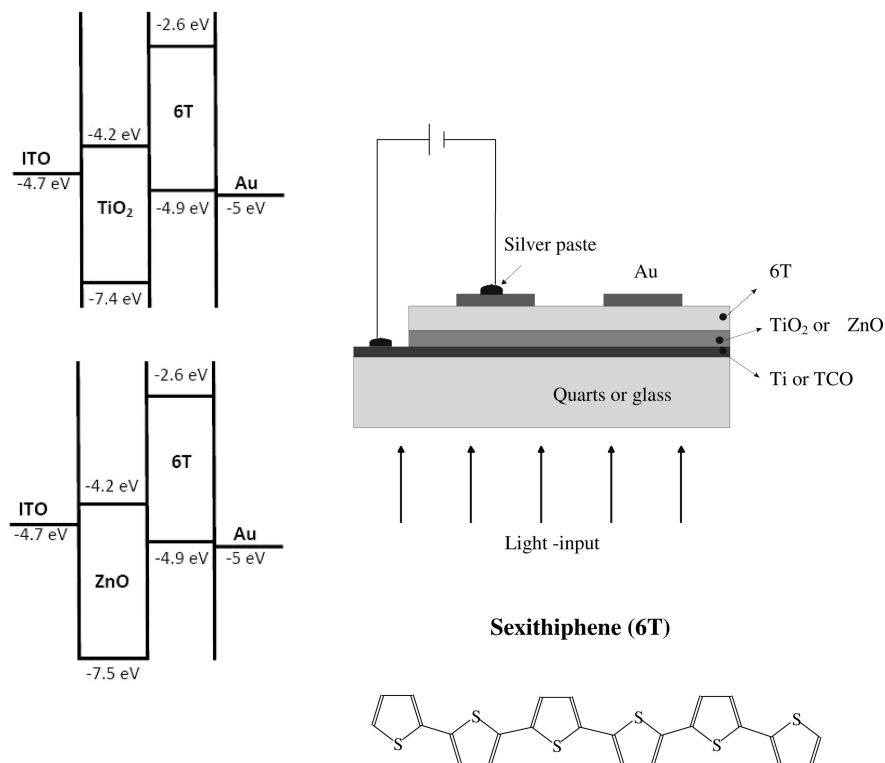


Figure 1. Schematic diagram of the samples seen in cross-section (right). For clarity, only two gold contacts are shown. Each sample includes 8–16 gold contacts. (Ti film and ITO are used as TCO in the case of e-beam and spray-pyrolysis TiO₂ samples, respectively). The Au electrode is positive vs the TCO back electrode. Energy level diagrams for Au/6T/TiO₂/ITO and Au/6T/ZnO/ITO. These are the flat-band situation (left).^{13,16,17}

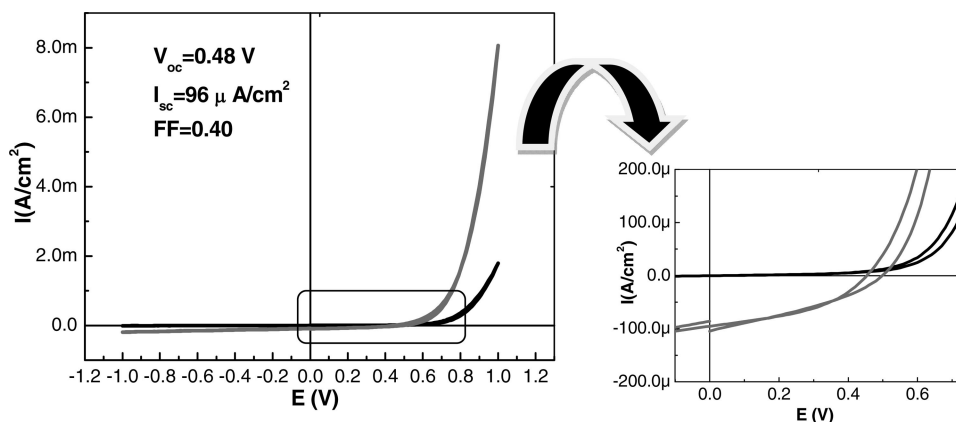


Figure 2. (Left) I – V curves of a Au/6T/TiO₂/TCO/glass device in the dark (black) and under 480 nm, 5 mW irradiation (gray). In the forward direction (positive voltage) the gold is positive. (Right) Enlargement of the left panel.

thoroughly cleaned in an ultrasonic bath with ethanol and acetone and dried with dry air before deposition of TiO₂. One hundred nanometer thick e-beam deposited TiO₂ substrates were received from Dr. L. Slooff at ECN (Energy Research Centre of The Netherlands).

Thin films of 6T were deposited using a thermal-evaporation setup.¹³ Sexithiophene powder is obtained from Syncom (Groningen, The Netherlands) and purified by zone-sublimation (courtesy of D. Fichou, Laboratoire des Matériaux Moléculaires, CEA Saclay, Thiais, France). On top of this oligomer film gold contacts were evaporated. Construction of the cell is schematically presented in Figure 1.

Measurements. The current–voltage (I – V) characteristics of 6T/TiO₂-based cells were examined with a Keithley model 2400 source meter in vacuum (10^{-3} mbar) and nitrogen ambient.

A continuous 20W power Nd:YVO₄ laser (Spectra Physics, millennia), operating at 532 nm, was used to irradiate the films.

Neutral-density filters were used to reduce the laser power to 0.02 mW. The luminescence is detected with a LN-cooled CCD camera (Princeton Instruments). Corrections for the filters and sensitivity of the CCD camera are always applied.

Electrochemical impedance spectroscopy (EIS) was used to characterize the Au/6T/TiO₂/TCO/glass and Au/6T/ZnO/TCO/glass devices in vacuum, air, or nitrogen in the dark at room temperature from -2.4 to 2.4 V. An EG&G 283 Potentiostat and a Frequency-Response Analyzer (Schlumberger Solartron 1255) were used to measure the I – V characteristics, the C – V curves, and the impedance spectra as a function of a dc bias in the 1 – 10^6 Hz frequency range. The amplitude of the applied voltage is 10 mV. A Pfeiffer vacuum system, consisting of a diaphragm prevacuum pump combined with a turbo molecular pump and a pressure gauge, maintained the vacuum in the cryostat at 10^{-3} mbar.

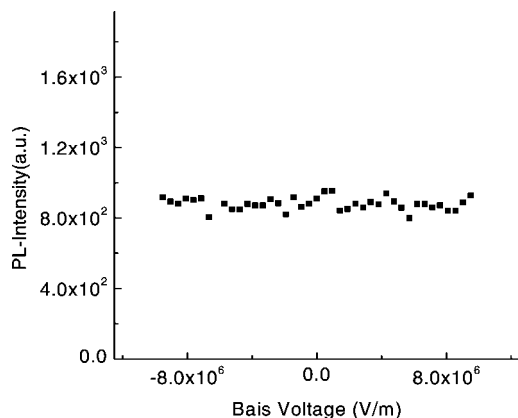


Figure 3. Electric field effect on the PL intensity of 6T (Au/210 nm 6T/100 nm TiO₂ (e-beam)/Ti/quartz) at 2.1 eV illumination for different dc applied bias voltages in V/m. The Au electrode is positive vs the Ti film back electrode.

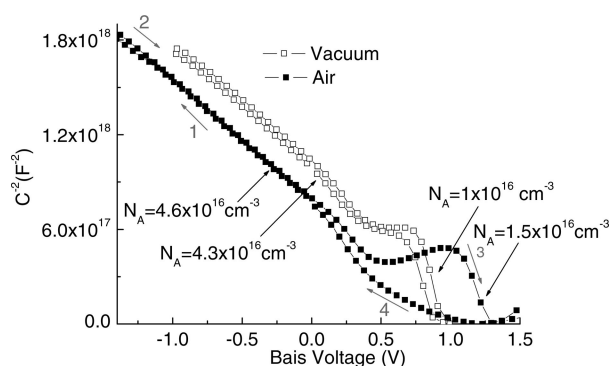


Figure 4. Mott-Schottky plots of an Au/210 nm 6T/100nm TiO₂ (e-beam)/Ti/quartz device in air and vacuum. The plots are recorded at 1 MHz with an ac signal amplitude of 10 mV. The Au electrode is positive vs the Ti film back electrode.

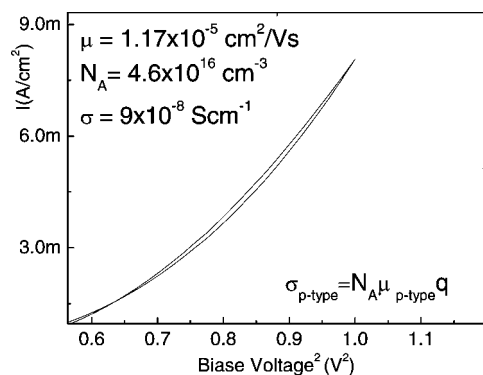


Figure 5. Current density– V^2 characteristic of an Au/210 nm 6T/100nm TiO₂ (e-beam)/Ti/quartz device in air. The Au electrode is positive vs the Ti film back electrode.

Results and Discussion

The I – V curves of Au/6T/TiO₂/TCO/glass device in the dark and upon 480 nm and 5 mW illumination are shown in Figure 2. In the forward direction (positive voltage) the gold is positive biased with respect to the TCO contact. The device in the dark shows excellent diode-like characteristics. When illuminated with 480 nm light a photovoltaic effect occurs. As can be seen in the right-hand panel of Figure 2, V_{oc} , I_{oc} , and Fill Factor (FF) are calculated as 0.48 V, 96 μ A/cm², and 0.40, respectively.

In order to distinguish between neutral excitons and polarons, the electric field effect on the intensity of the photoluminescence (PL) of a 210 nm thick 6T film deposited on TiO₂ (e-beam) at

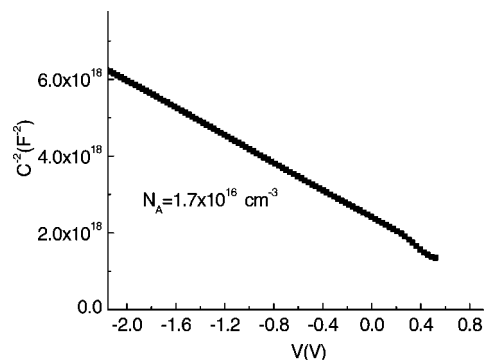


Figure 6. Mott-Schottky plot of Au/210 nm 6T/100 nm TiO₂ (e-beam)/Ti/glass from 0.5 to –2 V in an N₂ atmosphere. The Mott-Schottky plot is recorded at 1 MHz with an ac signal amplitude of 10 mV. The Au electrode is positive vs the Ti film back electrode.

2.1 eV illumination at different applied bias voltages is investigated. The results are presented in Figure 3. It is found that the PL intensity is independent of the applied dc bias voltage. This indicates that neutral excitons are involved in the charge transfer in the 6T films, which supports the conclusions by Fichou and Horowitz.²⁸

The donor–acceptor density in the p–n junction can be derived from Mott-Schottky analysis. For a p–n junction the following equation holds

$$C_{SC}^{-2} = \frac{2(\epsilon_A N_A + \epsilon_B N_B)}{\epsilon \epsilon_0 (\epsilon_A N_A \epsilon_B N_B) A^2} \left(V - \Phi_{fb} - \frac{kT}{e} \right) \quad (1)$$

in which ϵ_0 is the permittivity of a vacuum, ϵ_A and ϵ_B are the dielectric constants of the two involved materials, N_A and N_B the related donor and acceptor densities, A is the junction area, V is the applied bias voltage, and Φ_{fb} is the flat-band potential. If the product $\epsilon_B N_B$ is much larger than $\epsilon_A N_A$ material B drops out of the equation. In that case the $\epsilon_A N_A$ product can be determined from the slope of a C_{sc}^{-2} vs V plot. As will be pointed out below, indeed the $\epsilon_B N_B$ value related to TiO₂ and ZnO are larger than that of 6T, which allows us to disregard the contribution of the metal oxide film to the junction capacitance.

The space charge in the depletion region is determined by occupation of donor and acceptor states. The capture and emission of charge carriers in these states changes the space charge, which results in a capacitance change. Figure 4 shows the Mott-Schottky plots, i.e., $1/C^2$ vs V , of Au/210 nm 6T/100 nm TiO₂ (e-beam)/Ti/quartz device recorded with air and vacuum as ambient. For 6T, $\epsilon = 4$ and the contact diameter is 2 mm. A linear extrapolation to the potential axis gives an intercept of

$$V_0 = \Phi_{fb} + \frac{kT}{e} \quad (2)$$

from which a value for the flat-band potential, Φ_{fb} , can be derived.²³ The Mott-Schottky plots in air and vacuum show a difference. Flat-band potentials or built-in potentials of 1 (white squares) and ~ 1.25 V (black squares) are found, respectively. Close to the flat-band potential large differences are found between the graphs recorded in vacuum and air, indicating the active involvement of oxygen in the electrical properties of 6T. For small reverse voltages, the deep acceptor states are filled and do not contribute to the space charge. Accordingly, the space charge is only affected by shallow traps acceptor states, which leads to an acceptor density of $N_A = 1.5 \times 10^{16}$ cm^{–3} (for 6T this implies about 10 acceptors per million 6T molecules). For

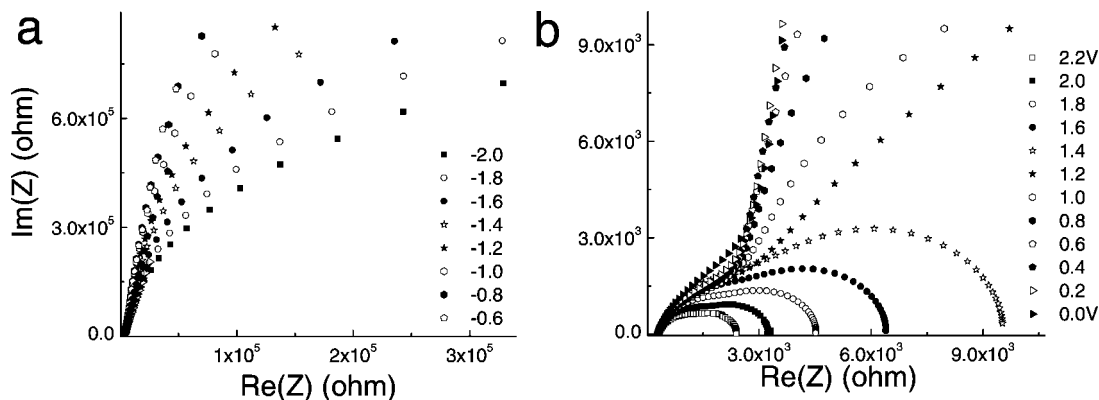


Figure 7. Impedance spectra of Au/160nm 6T/TiO₂ (SPD)/TCO/glass for negative voltages from -0.6 to -2.0 V (a) and positive voltages from 0 to $+2.2$ V (b) in the $1-10^6$ Hz frequency range. The Au electrode is positive vs the TCO back electrode.

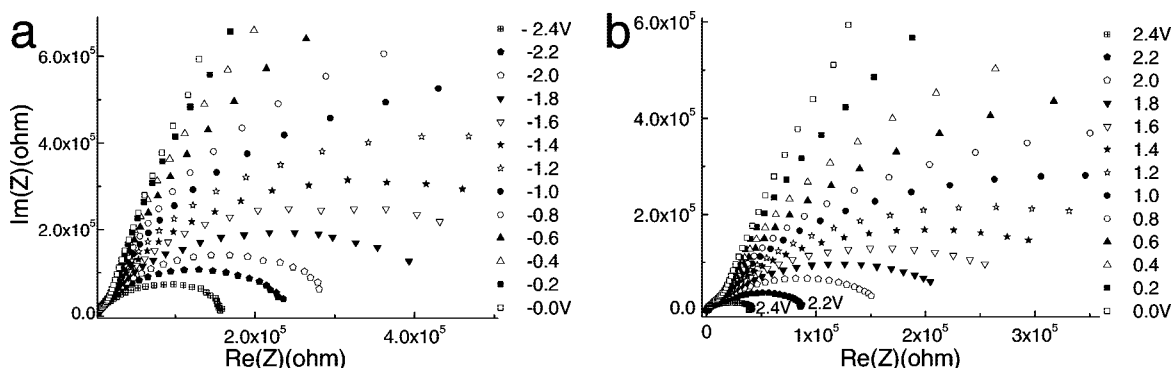


Figure 8. Impedance spectra of Au/160nm 6T/ZnO (SPD)/TCO/glass for negative (a) and positive (b) voltages from 0 to $+2.4$ V and in the $1-10^6$ Hz frequency range. The Au electrode is positive vs the TCO back electrode.

large reverse voltages both shallow and deep acceptors contribute to the space charge, and then the slope is much smaller, leading to $N_A = 4.6 \times 10^{16} \text{ cm}^{-3}$. In this regime there are two contributions to the space charge. The acceptor density obtained for small reverse voltages will be used to calculate the p-type conductivity of 6T in an Au/210 nm 6T/100nm TiO₂ (e-beam)/Ti/quartz device. Therefore, the mobility of the holes needs to be determined. The donor density of TiO₂ is reported to be approximately 10^{16} cm^{-3} in a single semiconductor Schottky barrier investigation.²² In combination with the dielectric constant of 50 for anatase the $\epsilon_B N_B$ value related to TiO₂ is $5 \times 10^{17} \text{ cm}^{-3}$, which exceeds the $\epsilon_A N_A$ product related to 6T by an order of magnitude, which justifies the used approximation of the Mott–Schottky equation.

The mobility of holes in sexithiophene (6T) is studied by measuring $I-V$ characteristics. 6T is vacuum evaporated in the temperature range $280-300^\circ\text{C}$ with no degradation or decomposition of the compound. The hole mobility of the charge carrier can easily be determined from the slope of a current density– V^2 plot (Figure 4) using Child's law²⁴

$$I = \frac{9}{8} \left(\frac{\epsilon_0 \epsilon_L}{L^3} \right) \mu V^2 \quad (3)$$

where μ is the mobility, V the applied voltage, and L the thickness of 6T. A hole mobility of $1.17 \times 10^{-5} \text{ cm}^2/\text{V s}$ is found. From this value and the acceptor density the p-type conductivity is calculated to be $9 \times 10^{-8} \text{ S cm}^{-1}$.

The acceptor density of 6T in an N₂ atmosphere is determined using the slope of the Mott–Schottky plot presented in Figures 5 and 6 for 210 nm 6T films on 100 nm TiO₂ (e-beam). The acceptor density is about $1.7 \times 10^{16} \text{ cm}^{-3}$ from the slopes of

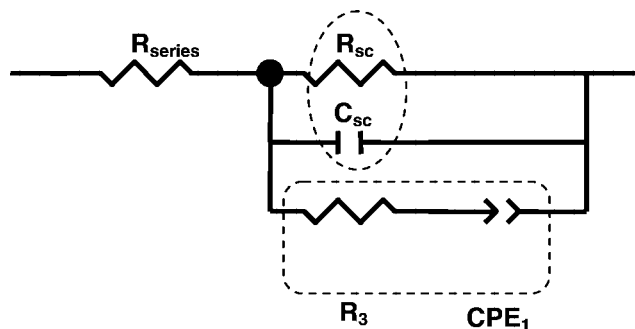


Figure 9. Equivalent circuit used for modeling Au/6T/TiO₂ (SPD)/TCO/glass and Au/6T/ZnO (SPD)/TCO/glass. Data were fitted using the Zview 2.1 software ($R_1 = R_{\text{series}}$, $R_2 = R_{\text{sc}}$ and $C_1 = C_{\text{sc}}$).

Figure 6, and this value compares very well with the acceptor density in air as ambient.

An EIS study has been performed of the charge transport across the TiO₂/6T and the ZnO/6T interfaces, as deposited by the SPD technique. The ac impedance spectra were recorded for Au/160 nm 6T/TiO₂/TCO/glass and Au/160 nm 6T/ZnO/TCO/glass cells in N₂ and the dark at room temperature as a function of a dc bias from -2.4 to $+2.4$ V. A typical impedance plot at 0.0 V is shown in Figure 7 for 6T/TiO₂ and Figure 8 for 6T/ZnO interfaces. The impedance spectra of the 6T film comprise a semicircle and combination of semicircles, which are depressed, as shown in Figure 7 for the TiO₂ substrate and Figure 8 for the ZnO substrate.

Figure 9 shows the equivalent circuit used for modeling the ac response of the Au/6T/TiO₂/TCO/glass and Au/6T/ZnO/TCO/glass devices, which consists of a resistor R_1 in series with a parallel combination of $R_2 C_1$, which is in parallel with a series combination of R_3 CPE₁. It is assumed that for 6T films

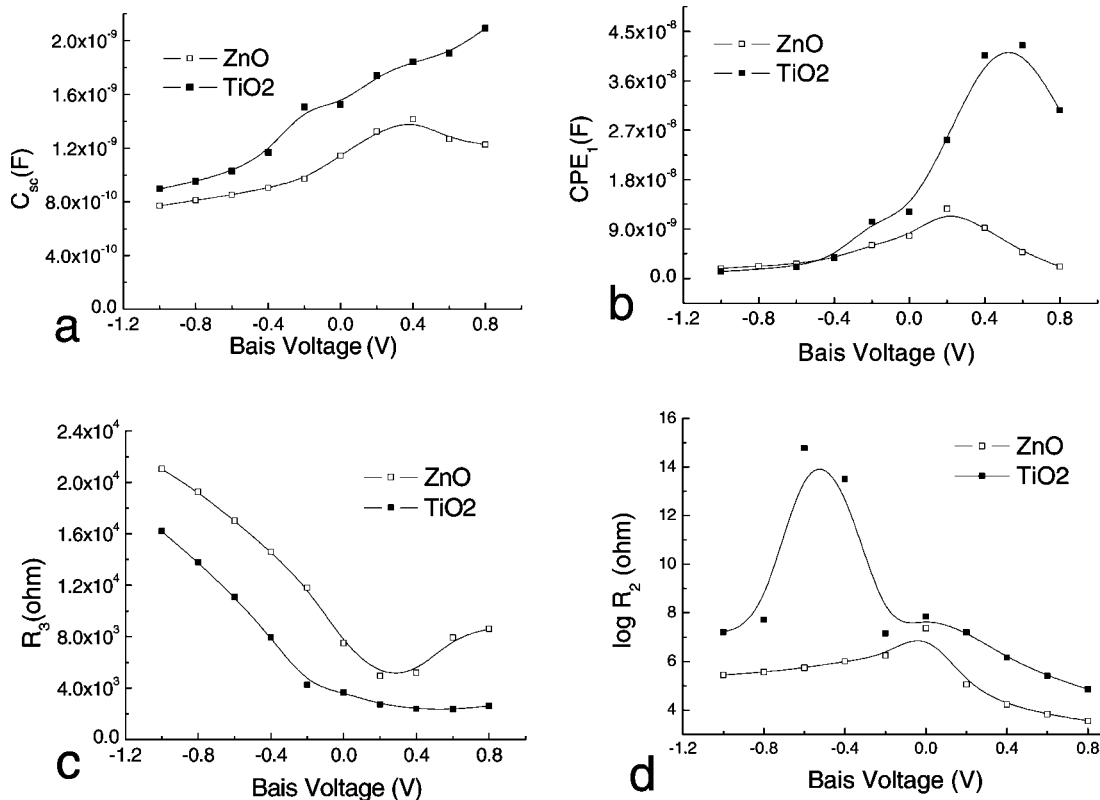


Figure 10. Values of (a) C_{sc} (F), (b) CPE_1 , (c) R_3 , and (d) R_2 (or R_{sc}) plotted versus applied dc bias. The Au electrode is positive vs the TCO back electrode.

deposited on TiO₂ and ZnO the relaxation time $\tau = R_{sc}C_{sc}$ represents the smallest time constant in the equivalent circuit. Then at very high frequencies, the system will only behave as a resistor because at very high frequencies both C and CPE are short circuited as $Z_C = 1/i\omega C$ and $Z_{CPE} = k_{CPE}/(i\omega)^\alpha$ with $\alpha \approx 0.8-1.0$. The TCO/TiO₂ (ZnO) interfaces are supposed to represent an ohmic contact described by a resistance.^{25,26} We also expect the contact of the Au/6T contact to be ohmic as the work function of gold is about 5.0 eV below the vacuum level and is approximately also the position of the valence band of 6T.^{13,16} The ohmic resistance at the TCO/TiO₂(ZnO) interface, the charge-transfer resistance at the TiO₂ (ZnO) contact, and the contact resistance between 6T and gold can be combined to an ohmic resistance ($R_1 = R_{series}$).

The necessity of a CPE instead of pure capacitance likely reflects the nonuniform nature of p-n heterojunctions. When α is close to unity, the k_{CPE} value extracted from a CPE element may be regarded as nearly equivalent to a frequency-dependent nonideal capacitance. The CPE_1 has an impedance $Z_{CPE} = k_{CPE}(i\omega)^{-\alpha}$, and for $\alpha > 0.8$, as obtained, in this study, the CPE_1 element can be regarded as nearly equivalent to a capacitance. It has been suggested by Goossens and Schoonman²³ that a well-fitted equivalent circuit representing both the space charge layer and the charge trapping has to comply with the following two requirements, i.e., (i) it must model the frequency response of the system accurately at every applied frequency and (ii) the equivalent circuit must be the same for every applied voltage with only the values of the circuit elements changing.

The time constant for charge trapping can be deduced from the elements of the equivalent circuit as $\tau = R_2C_1 = R_{sc}C_{sc}$. An average value of 8 μ s is calculated in the dc bias range, as shown in Figure 10. The R_3CPE_1 branch of the equivalent circuit is related to the presence of interface states at the 6T/TiO₂ interface. The shallow traps dominate the heterojunction time

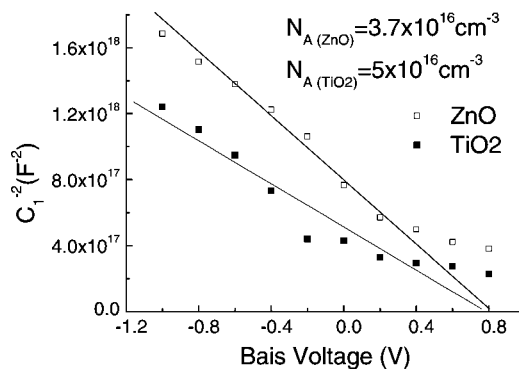


Figure 11. Mott-Schottky plot of 6T film-based devices Au/160 nm 6T/TiO₂ (SPD)/TCO/glass and Au/160 nm 6T/ZnO (SPD)/TCO/glass deduced from equivalent circuit data ($C_1 = C_{sc}$).

constant (parallel $R_{sc}C_{sc}$), while the deep traps effect the impedance measurements at low frequencies (series R_3CPE).²⁷ It is important to note that very good results are obtained in the whole dc bias range (from -1.2 to +0.8 V) for Au/6T/ZnO/TCO/glass. These results validate our proposed equivalent circuit.

As explained in the previous section (Figure 4), the space charge in the depletion region is changed by the capture and emission of charge carriers in deep and shallow acceptor states, which results in a capacitance change. The C_{sc} element of the equivalent in Figure 9 obeys the Mott-Schottky relation, as can be seen in Figure 11. The capacitances deduced from the equivalent circuit obtained by fitting the impedance spectra are compared with experimentally determined Mott-Schottky plots. The acceptor density of 6T/ZnO is found to be $3.7 \times 10^{16} \text{ cm}^{-3}$ and for 6T/TiO₂ $5 \times 10^{16} \text{ cm}^{-3}$ (as derived from the slope of the $C^{-2}-V$ plot). This value is in good agreement with the

previous value of $4.6 \times 10^{16} \text{ cm}^{-3}$ (Figure 4) for a 210 nm thick 6T film in 6T/100 nm TiO₂ (e-beam)/Ti/quartz devices.

Conclusions

In order to characterize the charge carrier dynamics, I – V , Mott–Schottky, and impedance measurements have been carried out on 6T/TiO₂ and 6T/ZnO organic–inorganic solar cells. From Mott–Schottky analysis of Au/6T/TiO₂/TCO/glass and Au/6T/ZnO/TCO/glass devices acceptor densities of 5×10^{16} and $3.7 \times 10^{16} \text{ cm}^{-3}$ are obtained for 6T, respectively. A p-type conductivity of $3 \times 10^{-8} \text{ S cm}^{-1}$ and a carrier mobility of $1.2 \times 10^{-5} \text{ cm}^2/\text{V s}$ for 6T on the TiO₂ sample are determined. The impedance data have been fitted in order to derive an electrical equivalent circuit. The $R_{sc}C_{sc}$ branch, which dominates in the high-frequency response, indicates shallow 6T-based traps, while the R_3CPE_1 , which dominates at lower frequencies, indicates deeper, interface-related states. The time constant for these elements has been found to be equal to 8 μs . This equivalent circuit has been validated in the potential range from -1.2 to $+0.8 \text{ V}$ for Au/6T/ZnO/TCO/glass.

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